

Kshitija Kumbharkar

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EDUCATION

University of Southern California — USC Viterbi

M.S. Engineering Management, GPA: 3.85

May 2026

Vishwakarma Institute of Technology, Pune

B.Tech Mechanical Engineering, GPA: 8.98/10

Jul 2022

EXPERIENCE

Product Intern, Product Strategy & Platform Roadmap

Jun 2025 – May 2026

Porsche Engineering Services (PEUS)

Carson, CA

- Led 0 → 1 development of Houston, an internal ADAS simulation platform, translating engineering and stakeholder requirements into a prioritized roadmap; scaled adoption from 10 to 50+ vehicles over a 12-month build.
- Built and launched Houston_CC, a geospatial module tracking 50,000+ km of ADAS fleet coverage; developed analytics for night-driving and lane-change calculations, reducing processing time from 4–5 hours to ~10 minutes.
- Managed A2D2 coverage KPIs across two program phases and built automation to improve engineering workflows, using program data to prioritize technical work and track progress.

Product Design Engineer, Systems Integration & PLM

Sep 2022 – Jul 2024

Mercedes-Benz Research & Development India

Pune, India

- Led wiring harness design for an EV supplier program across the product lifecycle, translating engineering requirements into release-ready designs using UG-NX, PREEVISION, and EPDM.
- Improved UG-NX design-data accuracy by 20% by identifying workflow inconsistencies and refining design specifications with engineering and supply-chain teams.
- Analyzed planning, production, and logistics KPIs and supported PPAP risk assessments and ISO workflow development, driving 30% lower operating costs, 15% higher process efficiency, and ~2% S-Class cost savings.

Design Intern, Operations & Stakeholder Management

Jan 2021 – May 2021

Dassault Systèmes

Pune, India

- Coordinated biweekly meetings across manufacturing, quality, and IT teams supporting the 3DEXPERIENCE platform, converting project issues and KPIs into actionable follow-ups.
- Improved project delivery time by 20% and client satisfaction by 15% through structured stakeholder coordination and KPI tracking.
- Supported cross-functional execution by maintaining project communication and aligning teams on priorities, dependencies, and next steps.

Supply Chain Intern, Process Automation & Workflow Design

Jun 2019 – Sep 2019

Cummins

Pune, India

- Designed automated assembly-line solutions for NHNT engine components and introduced min-max inventory controls, improving operational efficiency by 25%.
- Analyzed inventory patterns across facilities and adjusted replenishment parameters, reducing stockouts/overstock by 20%.
- Reduced inventory holding costs by 10% through process and inventory workflow improvements.

PROJECTS

Bluetooth Speaker — AI-Powered Bathroom Audio | B.Tech Capstone

- Designed a compact consumer-electronics enclosure around an audio PCB with USB-A, micro-USB, and 3.5 mm interfaces; aligned external ports to PCB datums using tolerance stack-up analysis.
- Developed retention bosses, mounting features, and enclosure interfaces in SolidWorks for prototype builds, iterating the design to accommodate packaging changes.
- Conceptualized a hands-free product experience combining voice control and AI-driven music personalization for a bathroom-focused Bluetooth speaker.

USC Housing Mobile Application — Engineering Project Management Capstone | USC ISE 515 | Spring 2025

- Developed the end-to-end project plan and secured faculty Board of Directors approval, defining the project charter, WBS, milestones, dependencies, resource allocation, and budget for a multi-phase rollout.
- Built a risk-management framework using probability-impact analysis, change management, and scope controls to identify and mitigate project risks.
- Structured execution around defined milestones and dependencies to support scope flexibility across project phases.

The Daily Brief — Multi-Model GenAI News Summarization App | USC ISE 547 | Apr 2026

- Led a 2-person team to build and deploy a 0 → 1 GenAI application using four LLMs, Flask, and Hugging Face Spaces across 120 source articles.
- Built an evaluation framework combining ROUGE-1, BERTScore, and human evaluation across five reviewers to compare model performance.
- Found that GPT-3.5 led automated ROUGE-1 performance by ~8%, while BART performed ~12% better in human evaluation, exposing a gap between automated and perceived quality.

TECHNICAL SKILLS

Engineering: SolidWorks, UG-NX, PREEVISION, EPDM, CATIA V5, CREO, ANSYS, Motor-CAD, PLM, tolerance stack-up, wiring harness design

Software & Data: Python, JavaScript, Node.js, SQL, Flask, GenAI/LLMs, geospatial systems, data analytics

Product & Program: Product development, requirements, roadmap planning, milestones, dependencies, risk analysis, KPI development, stakeholder management, process improvement, Jira, Confluence, Figma

LEADERSHIP

Viterbi Graduate Students Association — Senator, ISE Department | Aug 2025 – May 2026 Represented 150+ ISE students and organized industry events featuring senior leaders from Qualcomm, Western Digital, and FIS.